Minors and Cofactors of a Matrix

1. Introduction

Minors and cofactors are important concepts in linear algebra because they are used to find:

- Determinants of larger matrices
- The adjugate of a matrix
- The inverse of a matrix

The relationship is:

$$\text{Matrix} \rightarrow \text{Minors} \rightarrow \text{Cofactors} \rightarrow \text{Adjugate} \rightarrow \text{Inverse}$$

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2. Minor of an Element

Definition

The **minor** of an element of a matrix is the determinant obtained after deleting the **row and column** containing that element.

The minor of an element a_{ij} is written as:

$$M_{ij}$$

where:

- i represents the row number
- j represents the column number

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3. Finding Minors in a 2×2 Matrix

Consider:

$$A = \begin{bmatrix} a & b c & d \end{bmatrix}$$

Minor of a (M_{11})

Remove row 1 and column 1:

$$M_{11} = |d|$$

Therefore:

$$M_{11} = d$$

Minor of b (M_{12})

Remove row 1 and column 2:

$$M_{12} = |c|$$

Therefore:

$$M_{12} = c$$

Minor of c (M_{21})

Remove row 2 and column 1:

$$M_{21} = |b|$$

Therefore:

$$M_{21} = b$$

Minor of d (M_{22})

Remove row 2 and column 2:

$$M_{22} = |a|$$

Therefore:

$$M_{22} = a$$

So the matrix of minors is:

$$\begin{bmatrix} M_{11} & M_{12}\ M_{21} & M_{22} \end{bmatrix} ============= \begin{bmatrix} d & c\ b & a \end{bmatrix}$$

4. Example: Finding Minors of a 2×2 Matrix

Given:

$$A = \begin{bmatrix} 3 & 5 2 & 7 \end{bmatrix}$$

Find all minors.

Minor of 3

Remove row 1 and column 1:

$$M_{11} ====== |7|$$

Therefore:

$$M_{11} = 7$$

Minor of 5

Remove row 1 and column 2:

$$M_{12} ====== |2|$$

Therefore:

$$M_{12} = 2$$

Minor of 2

Remove row 2 and column 1:

$$M_{21} ====== |5|$$

Therefore:

$$M_{21} = 5$$

Minor of 7

Remove row 2 and column 2:

$$M_{22} ====== |3|$$

Therefore:

$$M_{22} = 3$$

The minor matrix is:

$$M = \begin{bmatrix} 7 & 2\,5 & 3 \end{bmatrix}$$

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5. Cofactor of an Element

Definition

A **cofactor** is the minor of an element multiplied by a sign factor.

The cofactor of a_{ij} is written as:

$$C_{ij}$$

Formula:

$$C_{ij} = (-1)^{i+j} M_{ij}$$

The sign depends on the position of the element.

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6. Cofactor Sign Pattern

For a matrix, the signs follow a checkerboard pattern:

$$\begin{bmatrix} + & - & + - & + & - + & - & + \end{bmatrix}$$

This means:

- C_{11} is positive

- C_{12} is negative
- C_{21} is negative
- C_{22} is positive

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7. Finding Cofactors of a 2×2 Matrix

Given:

$$A = \begin{bmatrix} a & b\, c & d \end{bmatrix}$$

The minors are:

$$\begin{bmatrix} d & c\, b & a \end{bmatrix}$$

Now apply the sign pattern:

$$C = \begin{bmatrix} +d & -c - b & +a \end{bmatrix}$$

Therefore, the cofactor matrix is:

$$C = \begin{bmatrix} d & -c - b & a \end{bmatrix}$$

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8. Example: Cofactor Matrix of a 2×2 Matrix

Given:

$$A = \begin{bmatrix} 3 & 5\, 2 & 7 \end{bmatrix}$$

From the previous example:

$$M = \begin{bmatrix} 7 & 2\, 5 & 3 \end{bmatrix}$$

Apply signs:

$$C = \begin{bmatrix} +7 & -2 - 5 & +3 \end{bmatrix}$$

Therefore:

$$\boxed{C = \begin{bmatrix} 7 & -2 - 5 & 3 \end{bmatrix}}$$

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9. Minors and Cofactors of a 3×3 Matrix

Consider:

$$A = \begin{bmatrix} a & b & c d & e & f g & h & i \end{bmatrix}$$

To find the minor of each element:

- Remove the element's row
- Remove the element's column
- Calculate the determinant of the remaining 2×2 matrix

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10. Example: Finding Minors of a 3×3 Matrix

Given:

$$A = \begin{bmatrix} 1 & 2 & 3 4 & 5 & 6 7 & 8 & 9 \end{bmatrix}$$

Find the minor of $a_{11} = 1$.

Remove row 1 and column 1:

$$M_{11} ====== \begin{vmatrix} 5 & 6 8 & 9 \end{vmatrix}$$

Calculate:

$$M_{11} = (5)(9) - (6)(8)$$

$$= 45 - 48$$

$$M_{11} = -3$$

Find the minor of $a_{12} = 2$.

Remove row 1 and column 2:

$$M_{12} ====== \begin{vmatrix} 4 & 6 7 & 9 \end{vmatrix}$$

Calculate:

$$M_{12} = (4)(9) - (6)(7)$$

$$= 36 - 42$$

$$M_{12} = -6$$

Find the minor of $a_{13} = 3$.

Remove row 1 and column 3:

$$M_{13} ====== \begin{vmatrix} 4 & 5\,7 & 8 \end{vmatrix}$$

Calculate:

$$M_{13} = (4)(8) - (5)(7)$$

$$= 32 - 35$$

$$M_{13} = -3$$

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11. Cofactor Matrix of a 3×3 Matrix

The cofactor matrix is obtained by applying signs:

$$\begin{bmatrix} + & - & +- & + & -+ & - & + \end{bmatrix}$$

For example:

$$C_{11} = +M_{11}$$

$$C_{12} = -M_{12}$$

$$C_{13} = +M_{13}$$

Using our values:

$$C_{11} = (-3)$$

$$C_{12} = -(-6) = 6$$

$$C_{13} = (-3)$$

The first row of the cofactor matrix is:

$$\begin{bmatrix} -3 & 6 & -3 \end{bmatrix}$$

The same process is repeated for all nine elements.

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12. Relationship Between Minors, Cofactors, and Determinants

The determinant of a matrix can be found using cofactors.

For:

$$A = \left[a \quad b \quad c\ d \quad e \quad f\ g \quad h \quad i\right]$$

Expanding along the first row:

$$\det(A) ======= aC_{11} + bC_{12} + cC_{13}$$

or:

$$\det(A) ======= aM_{11} - bM_{12} + cM_{13}$$

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13. Cofactor Matrix vs Adjugate Matrix

A common source of confusion:

Cofactor Matrix

The matrix containing all cofactors:

$$C = \left[C_{11} \quad C_{12}\ C_{21} \quad C_{22}\right]$$

Adjugate Matrix

The transpose of the cofactor matrix:

$$\mathrm{adj}(A) = C^T$$

Example:

If:

$$C = \begin{bmatrix} 7 & -2 - 5 & 3 \end{bmatrix}$$

then:

$$\text{adj}(A) ====================== \begin{bmatrix} 7 & -5 - 2 & 3 \end{bmatrix}$$

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14. Summary

Concept	Meaning	Formula
Minor	Determinant after removing row and column	M_{ij}
Cofactor	Signed minor	$C_{ij} = (-1)^{i+j} M_{ij}$
Cofactor Matrix	Matrix of all cofactors	C
Adjugate	Transpose of cofactor matrix	$\text{adj}(A) = C^T$

Key Points to Remember

1. A **minor** ignores signs and only calculates the smaller determinant.
2. A **cofactor** adds the correct positive or negative sign.
3. Cofactors are required to calculate determinants of larger matrices.
4. The adjugate is obtained by **transposing the cofactor matrix**.
5. The inverse of a matrix uses:

$$A^{-1} = \frac{\text{adj}(A)}{\det(A)}$$